Type MS · Explain how tooth decay happens and justify ways to prevent it

Answer 1 of 12 · Question 7 part b · 4 marks

2023 JUN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

Number				
7	(a) (i)	X = pulp (cavity); Y = dentine; Z = enamel;	Allow crown	3
	(ii)	• part Z/top surface/crown/enamel; • because it has ridges/crevices; • food becomes trapped/difficult to remove (with brushing);	Allow sugar for food	3
	(b)	• bacteria use/feed on food in teeth; • for respiration/energy; • (bacteria) produce (lactic) acid/lactate ; • (acid) dissolves enamel;		4
	(c)	• if had teeth could damage mother; • feed entirely from mother's milk/formula milk/babies are breastfed; • solid food not chewed/no solid food/liquid diet;		3
	(d)	• alcohol kills bacteria/reduces formation of plaque; • fluoride strengthens enamel/neutralises acid;		2

Total 15 marks

Type MS · Explain how tooth decay happens and justify ways to prevent it

Answer 1 of 12 · Question 7 part b · 4 marks

2023 JUN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

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Type MS · Explain how tooth decay happens and justify ways to prevent it

Answer 2 of 12 · Question 1 part a.iii · 3 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part a.iii in this mark scheme extract

1	(a)	(i)	molar;	Allow premolar	1
		(ii)	C, enamel		1
		(iii)	Any three from: • contains ridges/crevices; • larger surface area/more in contact with food; • food trapped; • provides nutrients for bacteria/reference to bacteria digesting/using food/producing acid; • more difficult to clean/reference to location in mouth (making tooth more difficult to reach);	Reject cusp	Max 3
	(b)	(i)	Any two from: • (both) decrease; • highest % of cases in 1983/lowest % cases in 2013; • (%) 15 year old children with tooth decay always higher/(%) 12 year old children with tooth decay lower; • (%) 15 year old children falls less overall/(%) 12 year old children falls more overall;		Max 2
		(ii)	clean/brush/floss teeth (more regularly / properly)/(more/regular) visits to dentist/ reference to fluoride in toothpaste/drinking water/less sugary foods in diet;	Allow better dental hygiene	1

Total question 1 = 8 marks

Type MS · Explain how tooth decay happens and justify ways to prevent it

Answer 2 of 12 · Question 1 part a.iii · 3 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how tooth decay happens and justify ways to prevent it

Answer 3 of 12 · Question 1 part b.ii · 2 marks

2020 JAN · Paper 02 · find part b.ii in this mark scheme extract

Number					
1	(a)	(i)	(tooth X) incisor(1) (tooth Y) canine(1)		2
		(ii)	grind/crush/chew food (1)		1
		iii)	Any two from: • Adult has premolars/wisdom teeth/3rd molar/more teeth/molars/(1) • adult teeth are permanent (1) • adult teeth are larger (1)	allow reverse argument for children	2
	(b)	(i)	London(1) less people visiting the dentist(1) OR North of England (1) More people visiting the dentist (as they have tooth decay) (1)	Area must be stated to award second mark	2
		(ii)	Any two from: • (molars) have ridges/crevices/grooves(1) • more likely that food can get trapped/larger surface area in contact with food/in contact with food for longer(1) • more difficult to reach with a toothbrush(1)	Ignore cusps	2

Total for Question 1 = 9 marks

Type MS · Explain how tooth decay happens and justify ways to prevent it

Answer 4 of 12 · Question 4 part c.i · 4 marks

2021 NOV · Paper 02 · extract 1 of 2 · find part c.i in this mark scheme extract

4 (a) (i)	• suitable scales; • axes labelled; • correct plots;;	Number of teeth on X axis, number of students on Y axis	4
(ii)	• 13; / 30 • x 100; • 43.3(%);	Allow 43(%)	3
(b)	• wisdom teeth/molars not developed; • jaw too small to accommodate extra teeth;		2
(c)(i)	• sugar coats/left in teeth/forms plaque; • bacteria use sugar (for respiration/energy); • convert sugar to lactic acid; • acid dissolves/erodes/breaks down enamel;	Allow feed on Allow release acid	4
(c)(ii)	any two from • regular brushing of teeth/use fluoride toothpaste/flossing; • visit dentist regularly; • add fluoride to tap water		2 Total 15

Type MS · Explain how tooth decay happens and justify ways to prevent it

Answer 4 of 12 · Question 4 part c.i · 4 marks

2021 NOV · Paper 02 · extract 2 of 2 · find part c.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how tooth decay happens and justify ways to prevent it

Answer 5 of 12 · Question 7 part a.ii · 3 marks

2023 JUN · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

Number				
7	(a) (i)	X = pulp (cavity); Y = dentine; Z = enamel;	Allow crown	3
	(ii)	• part Z/top surface/crown/enamel; • because it has ridges/crevices; • food becomes trapped/difficult to remove (with brushing);	Allow sugar for food	3
	(b)	• bacteria use/feed on food in teeth; • for respiration/energy; • (bacteria) produce (lactic) acid/lactate ; • (acid) dissolves enamel;		4
	(c)	• if had teeth could damage mother; • feed entirely from mother's milk/formula milk/babies are breastfed; • solid food not chewed/no solid food/liquid diet;		3
	(d)	• alcohol kills bacteria/reduces formation of plaque; • fluoride strengthens enamel/neutralises acid;		2

Total 15 marks

Type MS · Explain how tooth decay happens and justify ways to prevent it

Answer 5 of 12 · Question 7 part a.ii · 3 marks

2023 JUN · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

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Type MS · Explain how tooth decay happens and justify ways to prevent it

Answer 6 of 12 · Question 3 part a.iii · 3 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a.iii in this mark scheme extract

3	(a)	(i)	molar/pre-molar;		1
		(ii)	A = enamel; B = dentine; C = pulp (cavity); D = nerve;	D- accept sensory neurone	4
		(iii)	• ridged/crevices/uneven surface/large surface/eq.; • food/sugar/plaque/eq. becomes stuck/trapped/build up/stays longer on the molar; • bacteria able to act on (trapped) food/sugar/plaque; • acid produced:		3
	(b)		• contains calcium; • phosphate; • strengthens enamel; • resistant to decay;	accept fluoride for mp1 or mp2	4
	(c)		• regular brushing; • flossing; • (regular) visits to dentist/check-up at dentist; • use fluoride toothpaste/add fluoride to drinking water; • reduce consumption of sugar		3
					Total = 15

Type MS · Explain how tooth decay happens and justify ways to prevent it

Answer 6 of 12 · Question 3 part a.iii · 3 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how tooth decay happens and justify ways to prevent it

Answer 7 of 12 · Question 1 part b.ii · 1 mark

2019 JUN · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

1	(a)	(i)	molar;	Allow premolar	1
		(ii)	C, enamel		1
		(iii)	Any three from: • contains ridges/crevices; • larger surface area/more in contact with food; • food trapped; • provides nutrients for bacteria/reference to bacteria digesting/using food/producing acid; • more difficult to clean/reference to location in mouth (making tooth more difficult to reach);	Reject cusp	Max 3
	(b)	(i)	Any two from: • (both) decrease; • highest % of cases in 1983/lowest % cases in 2013; • (%) 15 year old children with tooth decay always higher/(%) 12 year old children with tooth decay lower; • (%) 15 year old children falls less overall/(%) 12 year old children falls more overall;		Max 2
		(ii)	clean/brush/floss teeth (more regularly / properly)/(more/regular) visits to dentist/ reference to fluoride in toothpaste/drinking water/less sugary foods in diet;	Allow better dental hygiene	1

Total question 1 = 8 marks

Type MS · Explain how tooth decay happens and justify ways to prevent it

Answer 7 of 12 · Question 1 part b.ii · 1 mark

2019 JUN · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how tooth decay happens and justify ways to prevent it

Answer 8 of 12 · Question 1 part b.i · 2 marks

2020 JAN · Paper 02 · find part b.i in this mark scheme extract

Number					
1	(a)	(i)	(tooth X) incisor(1) (tooth Y) canine(1)		2
		(ii)	grind/crush/chew food (1)		1
		iii)	Any two from: • Adult has premolars/wisdom teeth/3rd molar/more teeth/molars/(1) • adult teeth are permanent (1) • adult teeth are larger (1)	allow reverse argument for children	2
	(b)	(i)	London(1) less people visiting the dentist(1) OR North of England (1) More people visiting the dentist (as they have tooth decay) (1)	Area must be stated to award second mark	2
		(ii)	Any two from: • (molars) have ridges/crevices/grooves(1) • more likely that food can get trapped/larger surface area in contact with food/in contact with food for longer(1) • more difficult to reach with a toothbrush(1)	Ignore cusps	2

Total for Question 1 = 9 marks

Type MS · Explain how tooth decay happens and justify ways to prevent it

Answer 9 of 12 · Question 4 part c.ii · 2 marks

2021 NOV · Paper 02 · extract 1 of 2 · find part c.ii in this mark scheme extract

4 (a) (i)	• suitable scales; • axes labelled; • correct plots;;	Number of teeth on X axis, number of students on Y axis	4
(ii)	• 13; / 30 • x 100; • 43.3(%);	Allow 43(%)	3
(b)	• wisdom teeth/molars not developed; • jaw too small to accommodate extra teeth;		2
(c)(i)	• sugar coats/left in teeth/forms plaque; • bacteria use sugar (for respiration/energy); • convert sugar to lactic acid; • acid dissolves/erodes/breaks down enamel;	Allow feed on Allow release acid	4
(c)(ii)	any two from • regular brushing of teeth/use fluoride toothpaste/flossing; • visit dentist regularly; • add fluoride to tap water		2 Total 15

Type MS · Explain how tooth decay happens and justify ways to prevent it

Answer 9 of 12 · Question 4 part c.ii · 2 marks

2021 NOV · Paper 02 · extract 2 of 2 · find part c.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how tooth decay happens and justify ways to prevent it

Answer 10 of 12 · Question 7 part d · 2 marks

2023 JUN · Paper 02 · extract 1 of 2 · find part d in this mark scheme extract

Number				
7	(a) (i)	X = pulp (cavity); Y = dentine; Z = enamel;	Allow crown	3
	(ii)	• part Z/top surface/crown/enamel; • because it has ridges/crevices; • food becomes trapped/difficult to remove (with brushing);	Allow sugar for food	3
	(b)	• bacteria use/feed on food in teeth; • for respiration/energy; • (bacteria) produce (lactic) acid/lactate ; • (acid) dissolves enamel;		4
	(c)	• if had teeth could damage mother; • feed entirely from mother's milk/formula milk/babies are breastfed; • solid food not chewed/no solid food/liquid diet;		3
	(d)	• alcohol kills bacteria/reduces formation of plaque; • fluoride strengthens enamel/neutralises acid;		2

Total 15 marks

Type MS · Explain how tooth decay happens and justify ways to prevent it

Answer 10 of 12 · Question 7 part d · 2 marks

2023 JUN · Paper 02 · extract 2 of 2 · find part d in this mark scheme extract

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Type MS · Explain how tooth decay happens and justify ways to prevent it

Answer 11 of 12 · Question 3 part b · 4 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

3	(a)	(i)	molar/pre-molar;		1
		(ii)	A = enamel; B = dentine; C = pulp (cavity); D = nerve;	D- accept sensory neurone	4
		(iii)	• ridged/crevices/uneven surface/large surface/eq.; • food/sugar/plaque/eq. becomes stuck/trapped/build up/stays longer on the molar; • bacteria able to act on (trapped) food/sugar/plaque; • acid produced:		3
	(b)		• contains calcium; • phosphate; • strengthens enamel; • resistant to decay;	accept fluoride for mp1 or mp2	4
	(c)		• regular brushing; • flossing; • (regular) visits to dentist/check-up at dentist; • use fluoride toothpaste/add fluoride to drinking water; • reduce consumption of sugar		3
					Total = 15

Type MS · Explain how tooth decay happens and justify ways to prevent it

Answer 11 of 12 · Question 3 part b · 4 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how tooth decay happens and justify ways to prevent it

Answer 12 of 12 · Question 3 part c · 3 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

3	(a)	(i)	molar/pre-molar;		1
		(ii)	A = enamel; B = dentine; C = pulp (cavity); D = nerve;	D- accept sensory neurone	4
		(iii)	• ridged/crevices/uneven surface/large surface/eq.; • food/sugar/plaque/eq. becomes stuck/trapped/build up/stays longer on the molar; • bacteria able to act on (trapped) food/sugar/plaque; • acid produced:		3
	(b)		• contains calcium; • phosphate; • strengthens enamel; • resistant to decay;	accept fluoride for mp1 or mp2	4
	(c)		• regular brushing; • flossing; • (regular) visits to dentist/check-up at dentist; • use fluoride toothpaste/add fluoride to drinking water; • reduce consumption of sugar		3
					Total = 15

Type MS · Explain how tooth decay happens and justify ways to prevent it

Answer 12 of 12 · Question 3 part c · 3 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks